Soham Patel

Software Engineer — Robotics, Simulation & Data Systems

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PROFILE

Software engineer with about 2.5 years of production experience in healthcare data and payment systems, now studying Computer Engineering at Paderborn University, focused on robotics and embedded systems. My project work sits between software and hardware: a MuJoCo digital twin of a fischertechnik factory cell built from its PLC step tables and verified against 56 checks; LIDAR SLAM with occupancy-grid mapping and pure-pursuit control, ported from a Python Webots controller into the browser; YOLOv8 exported to ONNX for on-device inference; diffusion-based anomaly detection on brain MRI, with the weak result stated plainly. M.Sc. Information Technology as Gold Medalist, Rank 1 of cohort (9.55/10).

PROFESSIONAL EXPERIENCE

03/2025 – 08/2025
Part-time

Teaching Assistant — Decision Modelling & Data Analysis
OTH Amberg-Weiden · Weiden, Germany · Supervisor: Prof. Dr. Simon J. Preis

- Designed and delivered the applied data-analysis strand of a Master's module to a cohort of 30: SQL, statistical modelling, classification, regression and model evaluation.
- Built and maintained the KNIME reference workflows used across the module, covering data cleaning, feature engineering and evaluation on real industrial datasets.

02/2023 – 02/2024
Full-time

Software Engineer — Data Systems
Altera Digital Health (Allscripts India LLP) · Pune, India / Remote

- Built ETL pipelines on Microsoft Azure for healthcare audit-log data; data retrieval became about 30% faster.
- Implemented the encryption, access logging and retention rules required for health data, enforced in code rather than by convention.
- Refactored legacy C#/.NET services and introduced pytest checks that run before each deployment.

C# · .NET · Microsoft Azure · Blob Storage · T-SQL / MSSQL · Azure DevOps · pytest · Git

08/2022 – 01/2023
Full-time

Python Backend Engineer
Electrum IT Solutions · Vadodara, India

- Built FastAPI and Django services for a payments platform handling about 2,000 transactions a day.
- Implemented JWT authentication and role-based access control across the service boundary.
- Introduced centralised logging for failed payments, which made debugging about 25% faster.

Python 3 · FastAPI · Django · PostgreSQL · JWT · REST · Linux

12/2021 – 04/2022
Internship

Full-Stack Engineering Intern
Promact Infotech Pvt. Ltd. · Vadodara, India

- Delivered an expense-management product end to end: Angular 8 front end, ASP.NET Core 5 API, 9-table normalised MSSQL schema, JWT auth with role-based access, WebSocket live balances.
- Designed a graph-theory debt-simplification algorithm reducing the transfers needed to settle a group by up to 33%.

Angular 8 · ASP.NET Core · MSSQL · WebSocket · JWT · Agile

EDUCATION

Current

M.Sc. Computer Engineering
Paderborn University · Paderborn, Germany

- In progress. Focus on robotics, embedded systems and the software that connects sensing to action.

10/2024 – 08/2026

Graduate Coursework — Artificial Intelligence for Industrial Applications
Ostbayerische Technische Hochschule Amberg-Weiden · Amberg, Germany

- 45 ECTS · GPA 2.0. Digital twins and Industry 4.0, computer vision, generative models, autonomous systems (Embedded Intelligence graded 1.3).

08/2020 – 05/2022 **M.Sc. Information Technology**
Charutar Vidya Mandal University · Gujarat, India
- CGPA 9.55/10. **Gold Medalist, Rank 1** of a graduating class of approximately 120.
- Software engineering, databases and distributed systems; thesis in applied machine learning.

06/2017 – 05/2020 **B.Sc. Computer Science**
Sardar Patel University · Gujarat, India
- CGPA 8.5/10. Algorithms and data structures, operating systems, networks.

SELECTED PROJECTS

10/2025 – 03/2026 **Physics Digital Twin — fischertechnik Training Factory**

- MuJoCo rigid-body simulation of all four stations of the fischertechnik 536634 cell, built from the manufacturer's PLC step tables, TON timer literals and datasheets rather than from photographs; verified against 56 PLC-derived checks with 1.2 mm RMS prediction.
- Earlier versions: a browser 3D twin driven by the real PLC process image (9 DI / 14 DO), and an MQTT/InfluxDB maintenance dashboard (85% failure-prediction accuracy). Live on Google Cloud Run.

MuJoCo · Python · Siemens SCL · MQTT · InfluxDB · Three.js · Docker

2026 **Unsupervised Brain-MRI Lesion Segmentation (BraTS-PEDs)**

- Patch diffusion in pixel space: a DDPM trained on healthy 96×96 patches only; at inference each patch is partially noised and reconstructed, and the residual localises the lesion, refined by a dense-CRF guided by T1c contrast.
- **Result: DICE 0.076 · AUROC 0.713** — the masks do not yet segment usefully; reported as a baseline. The earlier latent-diffusion version underperformed and is documented as a negative result.

PyTorch · DDPM / DDIM · dense-CRF · HuggingFace · CUDA · Docker

2025 **Autonomous Maze Navigation — LIDAR SLAM**

- Log-odds occupancy-grid mapping, frontier exploration, A* planning on an inflated obstacle radius, pure-pursuit steering; five Webots maze worlds, ported from the Python controller into the browser, where the port is covered by unit tests.

Python · Webots · NumPy · Three.js · TypeScript

2025 **Real-Time Scene Understanding**

- YOLOv8n-seg exported to ONNX and run under ONNX Runtime Web, so camera frames never leave the visitor's device; paired with a self-hosted vision-language model for narration, with a detector-only fallback when none is reachable.

YOLOv8 · ONNX Runtime Web · TypeScript · FastAPI · Docker

TECHNICAL SKILLS

Languages	Python 3 · C# / .NET · TypeScript · SQL (T-SQL, MSSQL, PostgreSQL, MySQL)
Backend & Cloud	FastAPI · Django · ASP.NET Core · REST · JWT · Microsoft Azure · Docker · CI/CD · Git · Linux · pytest
ML & Vision	PyTorch · Scikit-Learn · XGBoost · HuggingFace · YOLOv8 · ONNX Runtime · Diffusion models (DDPM/DDIM) · Anomaly detection
Industrial & Data	Digital twins · MQTT · OPC-UA · PLC process images · InfluxDB · ETL pipelines · SCADA time-series · KNIME
Robotics & 3D	Webots · SLAM (occupancy grids) · A* / pure pursuit · MuJoCo · Three.js

AWARDS, LANGUAGES & ADDITIONAL

2025 **Grade 1.0 (highest distinction)** — keynote, OTH AI Research Conference: *Brownian Bridge Diffusion Models for 3D Shape-to-Image Medical Image Synthesis*.

2022 **Gold Medal & University Rank 1** — M.Sc. IT, CVM University (CGPA 9.55/10).

Languages English C1 (IELTS 6.5) · German A2, in active study toward B1 · Gujarati native · Hindi fluent.

Department student representative (two years) · Football: university team captain.

Work authorisation Valid German residence permit · EU Blue Card eligible via a recognised M.Sc. · no sponsorship required · available from September 2026.